

ICCT Passenger Flow Tensor (T_{ijk})

Model Documentation

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International Council on Clean Transportation (ICCT)
in collaboration with the Potsdam Institute for Climate Impact Research (PIK)

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1. Executive summary

The T_{ijk} passenger flow tensor is a calibrated $12 \times 12 \times 12$ array of modelled annual air passenger flows at the granularity of the 12 RICE model regions. Each entry $T[i, j, k]$ represents the number of travellers originating in region i who transit through region j en route to destination k . The tensor is the missing calibration input for a Walrasian equilibrium model of aviation tax incidence being developed at the Potsdam Institute for Climate Impact Research (PIK).

Two data sources are combined: the ICCT segment-level emissions inventory for the 2019 baseline year, and the Llano et al. (2023) bilateral tourism panel — a 74-country panel of foreign visitors by residence, air mode, whose most recent vintage in the published dataset is 2018 (i.e. a one-year gap to the 2019 baseline). Routing behaviour is modelled with a two-stage multinomial logit — Stage A determines the direct-versus-one-stop split, and Stage B allocates one-stops across transit regions. Calibration uses both segment-volume and hub-share loss signals, and the tensor totals 784 million inter-region OD passengers plus 2,638 million intra-region OD (grand total 3,423 M) in the 2019 baseline.

2. Model scope

2.1 The 12 RICE regions

The tensor's dimensions correspond to the 12 RICE (Regional Integrated Climate–Economy) regions:

ID	Label	Countries included
1	USA	United States
2	EU	EU-27 + Norway + Iceland + Switzerland + Liechtenstein + United Kingdom
3	JPN	Japan
4	RUS	Russia
5	EUR	Eurasia: Ukraine, Kazakhstan, Georgia, Belarus, Azerbaijan, Armenia, Uzbekistan, Turkmenistan, Kyrgyzstan, Tajikistan, Moldova
6	CHN	China + Hong Kong + Macau + Taiwan
7	IND	India
8	MDE	Middle East: Turkey, UAE, Saudi Arabia, Israel, Egypt, Iran, Iraq, Jordan, Kuwait, Lebanon, Oman, Qatar, Syria, Yemen, Bahrain
9	AFR	Africa (all sub-Saharan + North Africa excluding Egypt)
10	LAM	Latin America: Mexico, Brazil, Argentina, Colombia, Chile, Peru + rest of Central and South America + Caribbean
11	OHI	Other High Income: Canada, Australia, New Zealand, South Korea, Singapore
12	OAS	Other Asia: Thailand, Indonesia, Malaysia, Philippines, Vietnam, Pakistan, Bangladesh, Myanmar, Cambodia, Sri Lanka + rest of Asia not elsewhere classified

2.2 Baseline year

The canonical baseline year is 2019 (pre-COVID). This choice satisfies three constraints simultaneously: (i) it matches the vintage of the Llano et al. panel to within one year (panel data ends at 2018); (ii) it matches Maertens et al. (2020) Table 3 hub transfer rates (also 2018

vintage); (iii) it avoids the Russian airspace closure that began in February 2022 and the still-incomplete China market recovery visible in 2023 data.

A 2023 baseline was tested in early pipeline development but ultimately abandoned in favor of 2019 for the reasons above.

2.3 Calibration frame versus delivery frame

The 12-region definitions above are the canonical Potsdam scheme. Internally, however, the calibration pipeline merges Canada into the USA region and redistributes it back to OHI as a post-processing step. This is because the aggregate OHI region — Canada + Australia + New Zealand + South Korea + Singapore — has structurally bimodal routing patterns (Canada's US-adjacent trans-Atlantic flows versus Asia-Pacific hub flows of the other four), and modelling them together distorts predictions. In the calibration frame, OHI effectively means Asia-Pacific.

3. Data inputs

3.1 Llano et al. (2023) bilateral tourism panel

A 74-country, 1995–2018 panel of bilateral visitor flows decomposed by transport mode. We use the total number of non-resident visitors via air (VFR_pf_a) at the 2018 vintage, aggregated to the 12 RICE regions. Units are absolute persons per year.

Three systematic issues require correction before use:

Air-share representation in the panel.

The panel's mode shares derive from UNWTO transport-mode data, whose coverage of outbound air share varies across origins. The 2018 panel reports CHN at 1.0% air share on outbound international trips, HKG at 4.1%, and a cluster of Eastern European origins (UKR 13%, CZE 24%, ROU 24%) below regional norms. For any origin with international air share below 0.20, we replace it with the median air share of countries in the same RICE region that pass the threshold, and rescale that origin's bilateral air vector by (imputed/observed) — preserving destination structure. 17 origins are imputed; the global air total moves from 862 M (raw) to about 1.05 B (imputed).

Country-panel scope.

Llano et al. covers 74 of about 200 countries. After aggregation to 12×12 the imputed panel sums to 422 M international air trips, versus an inventory-derived 786 M for 2019. The remaining scope difference is closed via RAS biproportional fitting (§5.3), which scales the panel to match inventory-derived row and column marginals. Countries outside the panel are represented implicitly through the RAS step; the residual scope difference concentrates on EU-arriving flows, AFR-departing/arriving, and OAS-departing/arriving.

Transoceanic corridors.

For the 51 symmetric region pairs with no land or sea alternative (EU↔USA, EU↔OAS, USA↔CHN, etc.), we substitute the all-mode visitor count for the air-mode count. This corrects a systematic deflation where, for example, UK→USA is reported with 32% air share when it should be close to 100%.

3.2 ICCT 2019 emissions inventory

Segment-level flight records — one row per origin airport × destination airport × airline × aircraft type — with passenger counts. The inventory is used three ways:

- As the source of $F_{\text{obs}}[i, k]$, the inventory-derived 12×12 OD matrix used as RAS marginals (built segment-by-segment, deducting connecting traffic at each segment's arrival airport using published transfer rates).
- As $V_{\text{obs}}[a, b]$, the directed inter-region segment volumes that feed the Stage B segment loss.
- As the disaggregation target for Level-4 segment validation (§7.4).

3.3 Airport transfer rates

Per-region transfer share τ_i and inter-region hub transfer volumes $H_{\text{obs_inter_i}}$ are aggregated from published hub statistics. Sources include the EU top-30 airports panel, Maertens et al. (2020) Table 2 (top-25 global hubs, 2018) and Table 3 (supplementary high-rate hubs), and airport-specific published rates from individual airport authorities.

The published rate τ_i counts every transit passenger at the hub, including those whose feeder or distributor leg is itself intra-region (e.g., LAX-ATL-MIA in ATL's transfer count). For the Stage B target we need only inter-region routing — both legs cross region boundaries. We decompose τ using each airport's inventory-derived intra-region segment share s_a . Under independence between feeder and distributor leg, the share of transfer passengers with both legs inter-region is $(1 - s^2)$, and with at least one intra-region leg is s^2 .

Aggregated to regions:

$$\begin{aligned}\tau_{\text{inter_i}} &\approx \tau_i \cdot (1 - s_i^2) && \text{— inter-region} \\ \tau_{\text{intra_i}} &\approx \tau_i \cdot s_i^2 && \text{— intra-region}\end{aligned}$$

The current panel comprises 69 hubs:

Region	τ	n_hubs
USA	0.357	11
EU	0.190	30
JPN	0.090	2
RUS	0.420	1
EUR	0.240	1
CHN	0.163	5
IND	0.120	2
MDE	0.575	5
AFR	0.487	2
LAM	0.347	3

OHI	0.148	4
OAS	0.116	3

3.4 Fare data

The fare signal used in the calibration is a seat-weighted average 2019 economy class fare per RICE region pair, tax-inclusive (purchased from RDC Aviation Ltd.). It is the input to both Stage A (fare_penalty_onestop feature) and Stage B (norm_fare weight) of the MNL. The RDC fare dataset covers 96 of 132 region pairs with ≥ 10 observations; the remaining pairs use a log-linear distance regression fitted on the same source as a fallback.

4. Methodology

4.1 Schedule aggregation and feasible itineraries

Flight schedules (embedded in the ICCT inventory) are aggregated into directed RICE region-pair legs with annual frequency, seat capacity, capacity-weighted distance, block time, and seat-weighted fare.

One-stop itineraries ($i \rightarrow j \rightarrow k$) are enumerated subject to:

- Both legs must operationally exist ($\text{freq}_{ij} > 0$ and $\text{freq}_{jk} > 0$).
- No backtracking — first-leg distance $<$ total distance.
- Total flown distance $\leq 1.30 \times$ the direct (i, k) capacity-weighted leg distance, using self-consistent capacity-weighted distances on both sides of the inequality.

Itinerary filtering could be more restrictive and mandate both flight legs to have identical or codeshare airline per an internal Codeshare Dictionary. However, since the T_{ijk} pipeline operates at course RICE region-pair level, the codeshare restriction was deemed unnecessary. A diagnostic test using 2023 inventory and codeshare dictionary finds that only 4.9% of predicted transit volume would be excluded under a strict filter, with most of the exclusion concentrated on RUS-transit routings attributable to post-2022 sanctions. The remaining exclusion is also arguably an overestimate: a growing share of passengers self-connect on separate tickets, so real-world one-stop flows include some itineraries that fail a strict airline codeshare test.

Connecting fares are constructed as summed segment fares scaled by a per-transit-region multiplier that encodes the nonstop premium that segment-level fares would miss: MDE 0.60 (Gulf sixth-freedom undercut), AFR 0.65 (Ethiopian ADD hub), OAS 0.70 (intra-Asia LCC + Singapore), default 0.80 (Brueckner 2003 alliance discount). One origin-specific override applies: (LAM $\rightarrow$ MDE) = 0.80, reflecting that non-Brazil LAM does not route via the Gulf in practice.

4.2 Two-stage routing model

Routing share $\text{share}[i, j, k]$ is decomposed into a binary direct-versus-one-stop logit (Stage A) and a conditional MNL over transit choice given one-stop (Stage B).

Stage A — binary logit on direct share with 6-vector θ :

$$\begin{aligned} \text{logit } P(\text{direct} \mid i, k) = & \theta_0 + \theta_1 \log_seats_direct + \theta_2 \log_freq_direct \\ & + \theta_3 \text{time_penalty_onestop} + \theta_4 \text{fare_penalty_onestop} \\ & + \theta_5 \log_dist_ik \end{aligned}$$

Stage B — conditional MNL over transit:

$$\begin{aligned} w(i \rightarrow j \rightarrow k) &= \exp(-\alpha \cdot \text{total_time}) \cdot \exp(-\beta \cdot \text{norm_fare}) \cdot \text{bottleneck_seats}^\gamma \\ \text{share_onestop}[i, j, k] &= w / \sum_j w \end{aligned}$$

norm_fare is the one-stop's fare divided by the cheapest one-stop fare in (i, k).

Combined:

$$\begin{aligned} \text{share}[i, i, k] &= P(\text{direct} \mid i, k) & (j=i \text{ sentinel} = \text{'direct'}) \\ \text{share}[i, j, k] &= (1 - P(\text{direct})) \cdot \text{share_onestop}[i, j, k] & \text{for } j \neq i, k \end{aligned}$$

4.3 F_obs construction

F_obs is built from the Llano et al. panel in three steps: (i) air-share imputation (§4.1), (ii) transoceanic all-mode override for physically-air-only corridors (§4.1), and (iii) RAS biproportional fitting to inventory-derived row and column marginals from the 2019 OD matrix. The 2019 OD matrix used as RAS reference is itself built from the segment-level inventory by applying a per-segment $(1 - \tau_{\text{arrival}})$ deduction at every airport pair — $\text{pax}(a \rightarrow b) \times (1 - \tau_b)$ — to strip out the connecting-pax fraction and estimate true origin-destination demand on that leg. These segment-level OD estimates are then aggregated to region pairs to form the row/column marginals. Both sides of the RAS therefore arrive at the fitting step already at OD scale: Llano et al. VFR_pf_a counts destination-country arrivals by residence (no transit double-counting by construction), and the inventory F_obs marginals have the per-airport τ deduction baked in upstream. This is why the RAS itself does not apply a further $(1 - \tau)$ at the region level — doing so would double-count the deduction, an issue that collapsed AFR/LAM/RUS row/col targets when high- τ Table 3 hubs (ADD, PTY, SVO) entered the panel in 2026-05.

Mass evolution: the pre-RAS imputed Llano et al. panel sums to 422 M inter-region trips; RAS scales it to 786 M to match inventory row/col sums. The diagonal is forced to zero, and off-diagonal seeds use a 1.0-passenger floor.

4.4 Tensor assembly and Canada split

The inter-region tensor is computed on the calibration frame as:

$$T[i, j, k] = F_{\text{obs}}[i, k] \cdot \text{share}[i, j, k]$$

In the calibration frame, Canada and the United States are combined into an artificial North America region to approximate real-world aviation market dynamics. Canadian flows are then redistributed back into OHI in the delivery frame using inventory-derived CA-vs-US shares per axis:

- Origin axis on outbound NAM: $p_{\text{origin_ca}}[k] = \text{share of NAM} \rightarrow k \text{ segment pax originating in Canada.}$
- Destination axis on inbound NAM: $p_{\text{dest_ca}}[i] = \text{share of } i \rightarrow \text{NAM inbound landing in Canada.}$

- Transit axis on true one-stops ($j = \text{USA}, j \neq i$): geometric-mean proxy $\sqrt{p_{\text{origin_ca}} \cdot p_{\text{dest_ca}}}$.
- Direct sentinels ($j = i$) follow the origin assignment.

Mass is exactly conserved across the split.

4.5 Intra-region OD diagonal

The intra-region OD ($T[i, i, i]$) is the largest single component of the tensor — roughly 2.64 B passengers across the 12 diagonal cells, versus 0.77 B inter-region OD.

For each region i :

$$T[i, i, i] = \text{inventory_intra_seg}[i] \cdot (1 - \tau_i)$$

The $(1 - \tau)$ deflation backs out segment passengers to OD passengers by removing connecting traffic counted twice in segment volume. Raw τ is used here (not τ_{inter}) because ALL transit at intra-region hubs — whether the onward leg is intra- or inter-region — must be subtracted to yield true intra-region OD demand. This is the opposite of the Stage B use case (§4.3) where τ_{inter} is needed to remove only inter-region transit.

Per-region intra-region OD (millions of passengers, 2019):

Region	τ	Intra-region OD (M)	Share of intra total
USA	0.357	527.3	20.0%
EU	0.190	632.9	24.0%
JPN	0.090	100.1	3.8%
RUS	0.420	37.4	1.4%
EUR	0.240	6.1	0.2%
CHN	0.163	522.3	19.8%
IND	0.120	130.2	4.9%
MDE	0.575	86.6	3.3%
AFR	0.487	38.1	1.4%
LAM	0.347	194.8	7.4%
OHI	0.148	131.8	5.0%
OAS	0.116	230.2	8.7%
Total	—	2,637.9	100%

5. Calibration

θ is fit first, then (α, γ) — β is held fixed at 3.0 throughout.

5.1 Stage A target with V/F deflation

For each (i, k) with both direct and one-stop alternatives and positive F_{obs} :

$$\text{target_p}[i, k] = V_{\text{obs}}[i, k] \cdot (1 - \tau_{\text{inter_i}})(1 - \tau_{\text{inter_k}}) / F_{\text{obs}}[i, k]$$

clipped to $[0.05, 0.95]$.

$V_{\text{obs}}[i, k]$ is segment passengers on the $i \rightarrow k$ nonstop leg (which includes inter-region transit passengers using the leg). The $(1 - \tau_{\text{inter_i}})(1 - \tau_{\text{inter_k}})$ factor backs out direct OD by

removing the share of segment passengers who are inter-region transit at either endpoint. The undeformed V/F target has median 0.93 and 47% of pairs exceed 1.0 — meaning segment passengers exceed OD demand for nearly half of markets — which without deflation biases Stage A toward a direct-dominant solution.

Fitting is by weighted least squares in log-odds space (weights = F_{obs}).

Stage A fit: $\theta = (6.163, -2.090, 2.878, 0.921, -0.848, -0.221)$; loss = 1.115; $n_{\text{fit}} = 72$ markets.

5.2 Stage B loss

Segment volumes alone leave α and γ pinned to bounds because they only identify within-one-stop substitution, not the direct-versus-one-stop split. The Stage B loss therefore blends two signals with equal weightings:

- Segment loss — mean squared log-difference between predicted and observed directed inter-region leg volumes V_{obs} , restricted to pairs with $V_{\text{obs}} \geq 1000$.
- Hub-share loss — per covered region j , predicted transfer volume $\sum F_{\text{obs}}[i, k] \cdot \text{share}[i, j, k]$ compared to $H_{\text{obs_inter}}$ share across covered regions in log-share space. Comparing shares (not levels) cancels the hub-coverage bias.

Stage B fit: $\alpha = 0.005$ (lower bound), $\gamma = 0.757$. Blended loss = 1.229 (segment = 1.560, hub-share = 0.897).

5.3 β fixed at 0.6

β is fixed at 0.6 rather than fitted jointly with (α, γ) . Free-fit β is not viable: fare and total-time features are strongly collinear across the one-stop itinerary set (Pearson $r = 0.955$ pooled; within-market median $r \approx +0.68$), producing a ridge in the (α, β) likelihood surface that the optimizer resolves by pushing β to whichever bound it walks into.

To pick a defensible fixed value, we anchored the sensitivity grid to the airline-itinerary-choice MNL literature range. Lurkin et al. (2017) report uncorrected fare coefficients of -0.0025 to -0.0049 per dollar (high-yield versus low-yield segments), corresponding to $\beta \in [0.75, 1.47]$ in our normalisation with $\text{norm_fare} = \text{total_fare} / \text{min_fare_in_}(i, k)$ and typical min fare $\approx \$300$; endogeneity-corrected estimates are 40–70% larger, spanning $\beta \in [1.08, 2.07]$. Two anchor points below this band ($\beta = 0.3, 0.6$) were added because the literature values are almost entirely estimated on US domestic short-to-medium haul markets, while the T_{ijk} tensor is dominated by long-haul international corridors where fare sensitivity is empirically lower: long-haul demand is a mix of business travellers with weakly elastic price response and leisure travellers who book further in advance under stable published fares, so a lower β is plausibly the right prior in this regime. One anchor above the band ($\beta = 3.0$) was retained to bracket the previous canonical value.

With (α, γ) refitted at each grid point, blended loss decreases monotonically as β falls ($1.42 \rightarrow 1.33 \rightarrow 1.26 \rightarrow 1.24 \rightarrow 1.23 \rightarrow 1.23$); $\beta = 0.3$ is the loss minimum within the swept range, with $\beta = 0.6$ essentially tied ($\Delta\text{loss} < 0.3\%$). We adopt $\beta = 0.6$ as the canonical value — the conservative choice that sits just below the uncorrected low edge ($\beta_{\text{dollar}} \approx -0.002$ per dollar; 80% of the

Lurkin high-yield uncorrected magnitude) rather than the sweep-minimum $\beta = 0.3$ ($\beta_{\text{dollar}} \approx -0.001$ per dollar; 40% of that edge). The haul-composition argument justifies sitting below the range but not by an arbitrary amount, and $\beta = 0.6$ keeps the model within a factor of ~ 1.25 of a directly reported literature value while capturing the haul-mix adjustment. The uncorrected literature band (rather than the endogeneity-corrected band) is the reference because our estimator, like Lurkin's before their instrument was applied, has no price-endogeneity correction.

Sensitivity analysis confirms that all top-20 tensor cells ($\sim 90\%$ of inter-region OD mass) are exactly invariant to β within $[0.3, 3.0]$; only the routing distribution across single-hub regions like RUS varies materially ($\approx 2.2\times$ across the swept range).

5.4 α at lower bound

α converges to its lower bound of 0.005/hr, far below the $-0.30/\text{hr}$ time coefficient reported by Lurkin et al. (2017) for a linear-time airline itinerary-choice MNL — the most directly comparable published value. To test whether α is genuinely near zero or pinned by a bound artefact, we held α at literature values corresponding to representative values of time ($\$15/\text{hr} \rightarrow \alpha = 0.15$, $\$25/\text{hr} \rightarrow \alpha = 0.25$, $\$40/\text{hr} \rightarrow \alpha = 0.40$) and re-optimised γ (with β held at its canonical fixed value; the qualitative pattern is invariant to reasonable β choices). Blended loss rises monotonically from 1.23 at the bound to above 2.0 at $\alpha = 0.4$, with both segment and hub-share losses worsening. Per-region transit rebalances toward geographically direct hubs (RUS transit grows from ≈ 1.3 M to ≈ 8 M at $\$25/\text{hr}$ VoT; EUR from ≈ 2.5 M to ≈ 4.4 M) and away from the empirically dominant sixth-freedom hubs (MDE drops from ≈ 27 M to ≈ 23 M). Forcing literature-consistent time sensitivity pulls the model in the wrong direction relative to the observed hub-share targets.

The literature $H_{\text{obs_inter}}$ targets show MDE at $\text{lit_}\tau_{\text{inter}} = 0.51$ (highest of any region), meaning that in the observed data real routing accepts substantial geographic detour when carriers offer competitive frequency, price, and schedule alignment. This structural signal survives in the calibration only when α is small; forcing α up erodes it. The $\alpha = 0.005$ outcome is therefore what the joint calibration objective requires given the data available at this aggregation scale, not evidence that travellers do not value time.

The implied value of time at the canonical $(\alpha, \beta) = (0.005, 0.6)$ is roughly $\$2.50/\text{hr}$ — an order of magnitude below leisure literature ($\$10\text{--}25/\text{hr}$) and two orders of magnitude below business estimates ($\$30\text{--}60/\text{hr}$). This is best read as a signature of the aggregation scale rather than a preference-parameter estimate. At region-pair granularity one traveller's fast-preference and another's cheap-preference average out; the residual signal is dominated by capacity (which is why γ identifies cleanly) and by fare only weakly. Any fixed β leaves the implied VoT dominated by α — and α at this scale is driven toward zero by the calibration objective regardless of β . Recovering a literature-consistent VoT would require both a larger α ($\approx 0.30/\text{hr}$ per Lurkin et al. 2017) and per-carrier fare data to break the α – β collinearity; neither is achievable at region-pair aggregation without breaking the routing fit. Value of time for downstream economic models should therefore be sourced from the published literature rather than back-solved from our fitted α .

6. Validation

We do not validate against an external OD dataset — none is publicly available at global coverage. Validation consists of internal pipeline consistency checks (Levels 1 through 4) plus a qualitative web-search audit.

6.1 Level 1 — flow conservation

$T.\text{sum}(\text{axis}=1)[i, k] = F_obs[i, k]$ by construction. Verified numerically each run.

6.2 Level 2 — F_obs marginals versus regional totals

F_obs row totals (outbound) and column totals (inbound) for 2019, in millions of passengers:

Region	Out (M)	In (M)
USA	111.2	108.3
EU	153.0	145.0
JPN	28.9	27.9
RUS	17.5	15.9
EUR	18.2	16.3
CHN	46.2	46.7
IND	33.2	27.2
MDE	113.5	145.2
AFR	39.3	34.3
LAM	69.1	64.0
OHI	54.4	63.0
OAS	81.9	72.4

MDE's large inbound-outbound asymmetry (113 out, 145 in) is the sixth-freedom hub signature — Gulf carriers' net transit role appears here as inbound > outbound in the origin-region-of-segment aggregation.

6.3 Level 3 — predicted hub transfer rate versus literature τ_inter

L3 compares rates rather than volumes. The literature τ_inter defined in §4.3 is a hub-airport-weighted rate: the share of total dep+arr at the sampled hub airports that is inter-region transit. The model's natural counterpart at hub scope is:

$$\text{pred_}\tau_inter_hub[j] = \text{pred_transit}[j] / \text{total_hub_dep_arr}[j]$$

where $\text{pred_transit}[j] = \sum F_obs[i, k] \cdot \text{share}[i, j, k]$ for $i \neq j, k \neq j$.

Region	τ_inter (lit)	pred_ τ_inter_hub	ratio
USA	0.147	0.018	0.12
EU	0.085	0.020	0.23
JPN	0.054	0.053	0.98
RUS	0.215	0.044	0.20
EUR	0.240	0.154	0.64
CHN	0.091	0.020	0.22
IND	0.058	0.040	0.70
MDE	0.510	0.106	0.21
AFR	0.306	0.092	0.30
LAM	0.116	0.002	0.02

OHI	0.134	0.048	0.36
OAS	0.071	0.074	1.04

Interpretation:

- JPN and OAS are near 1:1, indicating well-fit transit hub roles.
- Multi-hub regions (USA, EU, MDE, CHN, AFR, OHI) under-predict in the 0.20–0.35 range. Model captures relative ordering but delivers 25–35% of literature τ_{inter} . This is the 'structural under-prediction' pattern that reflects (i) the 1.30 backtrack filter excluding some operationally rare but real long-haul routings, (ii) the calibration's segment loss pulling toward direct-flight assignments, and (iii) the literature τ_{inter} at hubs including connections that are only approximated at 12-region resolution.
- Single-hub regions (RUS, EUR, IND) fit plausibly at ratios of 0.40–0.65.
- LAM = 0.01 is a benchmark artefact rather than model failure: PTY's published τ largely reflects intra-LAM transit that is invisible at 12-region resolution.

6.4 Level 4 — segment-level RMSLE

The tensor is disaggregated to directed inter-region segment volumes $V[a, b]$ and redistributed to airport-pair segments within (a, b) proportional to seat capacity. Intra-region segments receive $\text{pred} = 0$ by construction (the model is inter-region only).

RMSLE is defined as $\sqrt{\text{mean}((\log_{10}(\text{pred}) - \log_{10}(\text{obs}))^2)}$.

Metric	All inter (n=14,897)	Inter top-quartile by V_{obs} (n=3,725)
Total observed pax	894 M	698 M
RMSLE	0.697	0.535
GEH<5 share	20.3%	6.4%
GEH<10 share	32.7%	10.2%
Ratio median	0.93	0.91
Ratio p10–p90	0.35 – 1.44	0.56 – 1.35

6.5 Level 5 — qualitative Google Flights audit

27 OD pairs across 12 representative region pairs were queried on Google Flights (default ranking). The audit confirms the operational transit set — which hubs appear as bookable connections for a given corridor — but cannot be read as a passenger-share estimate (Google's ranking is price plus convenience, not volume). Two vintage caveats: US-China direct collapsed post-COVID, and Russian airspace closed since February 2022, so any RUS transit on Europe-Asia or trans-Pacific routes is operationally impossible in the current schedule snapshot but was in service in 2019. The audit informed several calibration choices, including the LAM→MDE connecting-fare discount override.

7. Outputs

7.1 Tensor totals

- Total OD passengers (2019 baseline): 3,423 M
- Inter-region OD passengers: 784 M (includes ~18 M of US↔CA cross-border direct flows added at the intra-region step from inventory)
- Intra-region OD passengers (diagonal): 2,638 M
- Total predicted inter-region transit: 107 M

7.2 Nonstop share by corridor

Per (i, k), the model produces a direct-share matrix. Dense trunk pairs (USA↔EU, USA↔LAM, intra-Europe pairs) exhibit direct shares above 0.8. Long-haul pairs with strong hub competition (EU↔OAS, USA↔IND, AFR↔OAS) sit in the 0.2–0.5 range where routing is dominated by Gulf and Asian hubs. Full heatmap is provided as a figure in the accompanying materials.

7.3 Top transit hubs by corridor

Per (i, k), the dominant transit region varies as expected: Gulf carriers via MDE dominate EU↔Asia-Pacific destinations and USA↔IND corridors; OHI via SIN/ICN carries much of OAS↔USA and OAS↔EU; EU is the dominant transit for AFR↔non-Gulf destinations; OAS via SIN/BKK for IND↔OHI and CHN↔MDE/EU/USA.

8. Future work

The following items are areas where the model could be strengthened in a future release.

8.1 Data coverage extensions

Tourism panel's African coverage

The Llano et al. 2018 panel includes 5 of ~54 African countries (DZA, LBY, MAR, TUN, ZAF); most of West, East, and Central Africa are outside the panel. A next iteration could supplement the tourism panel with an auxiliary bilateral source (e.g. UN DESA migrant-stock × visitation rates) for major missing African origins.

RUS and EUR inventory coverage

The 2019 inventory used for F_{obs} marginals under-counts RUS↔EU segments (0.14 M observed versus an estimated 5 M+ reality) because Aeroflot / S7 / Rossiya trunks to EU capitals are absent from the provider's data, and Ukraine (KBP) is similarly under-counted.

8.2 Model specification refinements

Per-transit alternative-specific constants (ASC) in Stage A

The Stage A fare-penalty coefficient θ_4 currently carries an economically wrong sign because on nearly all high-volume corridors the cheapest connecting fare undercuts the direct fare (sixth-freedom / alliance discount) yet $P(\text{direct})$ remains high due to a nonstop premium. A diagnostic

replication with per-transit ASCs — the standard airline-itinerary-choice literature specification — reduces Stage A WLS loss by 18% and produces interpretable per-transit-region coefficients consistent with sixth-freedom hub economics. Downstream L4 validation metrics are essentially unchanged (RMSLE 0.706 versus 0.707), so this refinement is documented as a diagnostic but not adopted for v1.0. A future version aimed at a methods paper would benefit from adopting the ASC specification for interpretability.

Corridor-specific connecting-fare discounts

The current per-transit-region connecting-fare discount table (MDE 0.60, AFR 0.65, OAS 0.70, default 0.80) is qualitative midpoints from airline pricing literature. A future version could fit these as parameters against direct-share residuals on Gulf-touched corridors.

9. References

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